Humorized Computational Intelligence towards User-Adapted Systems with a Sense of Humor

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Abstract. This paper investigates the role of humor in non-task oriented (topic restriction free) human-computer dialogue, as well as the correlation between humor and emotions elicited by it in users. A joke-telling conversational system, constructed for the needs of this research, was evaluated by the users as better and more human-like than a baseline system without humor. Automatic emotive evaluation with the usage of an emotiveness analysis system showed that the system with humor elicited more emotions than the other one, and most of them (almost 80%) were positive. This shows that the presence of humor makes computers easier to familiarize with and simply makes users feel better. Therefore, humor should be taken into consideration in research on user-friendly applications, as it enhances the interaction between user and system. The results are discussed and our concept of a user-adapted humor-equipped system is presented.

Keywords: Humor, Human-Computer Interaction, emotions, conversational systems, computational humor.

1 What Makes Humans Humans?

To dispel the reader's fears at the very beginning – this section is not aimed at answering the age-long question about the nature of human beings. What we are going to do below, however, is to discuss some features of interaction between humans and introduce our research in which we examined their role in interactions with machines. The results described in this paper show that we took one step further towards Humanized Computational Intelligence.

1.1 Chatting as a Part of Interaction

We, humans, have developed an ability to communicate with each other through the usage of language. Most of us do that every day, with different purposes. Some of our conversations are clearly task-oriented, while others just concentrate on maintaining social relations, amusement and pleasure coming from interaction itself. Having no talking partner is commonly associated with loneliness, sadness, and generally bad emotions, with can be proved easily by putting the phrase "no one to talk to" into

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Google (or any other Internet search engine). It is not a coincidence that most psychological therapies are based on conversation. However, it is the opportunity to talk just for fun (to chat), without a particular purpose, that actually makes us feel that we are part of society.

1.2 Humor as a Positive Factor

We are far from promoting humor as a cure for everything – there are, however, many robust proofs for its beneficial influence on our life. To name only few of them, we know that humor is often used as a measure to cope with negative emotions and moods, such as stress [1], anxiety, or depression [2]. Research conducted by Vilaythong et al. showed that exposing people to humorous contents (like funny videos) increased their feeling of hopefulness and obviously made them feel better [3]. Proofs for social benefits of humor were also provided by Cook and Rice [4] – the results of their experiments showed that a sense of humor in another person increases the perceived benefits of a relationship. According to Sprecher and Regan [5], the sense of humor is also one of the main characteristics we use when choosing a partner, which means that we like to interact with people who use humor. Finally, we have the research of Mulkay [6], who proved that we tend to use jokes when discussing difficult subject with others, which leads to the conclusion that humor actually makes our conversation easier.

2 What Makes Computers More Human?

The issues of human-likeness described in section **1** have not been mentioned here without reason – both the ability to chat and to use humor are said to be applicable such areas as Human-Computer Interaction or Humanized Computational Intelligence.

It has been convincingly demonstrated that humans treat computers as if they were social actors. According to SRCT (Social Response to Communication Technologies) theory, people respond to computers using the same social attitudes and behaviors they apply to other people [7]. This leads us to the statement that – if computers are really treated (or going to be treated) in similar way to humans – the role of mere tools in our lives is not the only one to be played by them. If we approach them as to social actors, it is obvious that we expect them to behave in as natural a way as possible, and the interaction between us should then go smoothly. Therefore, if we chat with other people, we would also be willing to do it when communicating with computers, and if we frequently exploit the benefits of humor in the interaction with other humans, they should have the same effect in interaction with machines.

"Machine talking" itself is a subject of research in the field of NLP (Natural Language Processing) – however, even here non-task oriented (topic or task restriction-free) discourse is still an underestimated issue. Implementing humor into such non-task oriented systems is an even more neglected area, and we believe it is a subject worthy of more intensive research.

3 Pun-Telling Freely Talking System

In our research we decided to fill that gap. We started a project aiming to construct a funny, humor-equipped freely talking system. For the subject of this enterprise we chose puns – a genre of jokes which is based on features of the language (such as homophony) and thus is relatively computable with the current methods of NLP. The base language of our research is Japanese, as it is rich in homophonic phrases which gives vast possibilities of generating puns (in Japanese called *dajare*).

Having constructed a simple joke-telling conversational system (described in our previou works [8] - see section **4.2** for a description of the algorithm), we checked its performance in an evaluation experiment, in which users talked first with the system without humor, and then with a similar one with humor. Next, we analyzed the chat logs with our emotiveness analysis system to check if the presence of humor actually influenced the user's emotions. Our premise was that, if successful, the users' emotions towards the system with humor should be more positive than in the case of the non-humorous one. The results are described below.

4 Systems Used in This Research

In this research we used two talking systems: with and without humor. The latter is equipped with our pun generating engine (see section 4.2). To analyze the results, we used an emotiveness analysis system, which recognizes emotions in utterances (see section 4.4).

4.1 Freely Talking System

The baseline system in this research is a text-based chatterbot called "Modalin" (developed by Higuchi et al. [8]) which uses the Internet to extract word associations for interlocutor utterances and adds modality to generated responses. For example, given an user input phrase: "- *Nanika sukina tabemono aru*? (What food do you like?)", the system extracts a keyword: "*tabemono*" (food), checks its associations on the Internet, finds the word "*oishii*" (tasting good) as the closest one to the keyword, and uses it to generate a sentence. Then, it adds modality to the generated phrase, and outputs the response, which in this case is: - *Maa, tabemono-wa oishii desu*. (Well, food tastes good.).

In the evaluation experiment, human subjects assessed over 80% of the extracted associations as correct. The experiment also showed that adding modality to the system's response improved its results in all criteria.

4.2 Pun Generator

The pun generating engine PUNDA, developed in our previous research [9], is also based on the Internet. From a user utterance it extracts a base word and transforms it using Japanese pun phonetic generation patterns, to create a phonetic candidate list.

Then, it checks all candidates in the Goo search engine[1], and chooses the one with the highest hit rate, i.e. the most common word that sounds similar. Next, it uses the KWIC on WEB - online Keyword-in-context sentences database [10] – to find a sentence with the chosen word and extracts part of the sentence starting with the word.

Below we present an example of the system in action:

User: ***Jaa, issho ni eiga wo mi ni ikanai?***
(So, will you go to see a movie with me?)
Base word: *eiga* (a movie)
Pun candidate: *eiga* (glory)
KWIC Sentence part: *...eiga wo hokotta* (was glorious)
System's response: ***Eiga*** **(movie)** ***to ieba eiga*** **(glory)** ***wo hokotta.***
(Speaking of movies, it was really moving!)

If no candidate was found using the user's input, the system randomly selects a pun from our pun data base.

4.3 Freely Talking Joking System

The two algorithms described above were combined to create a talking system which tells jokes. The system was named "Pundalin". As far as joke timing is concerned, we decided to apply a very simple rule – in every third turn of the conversation, the normal talking system's output was replaced by a joke, generated by the joking system. In other words, the user's every third utterance becomes an input for the joke generator, which generates an appropriate pun for it. Preliminary tests showed that joking in every third turn is optimal for this experiment. The system has also been described in our previous work [9].

4.4 Emotiveness Analysis System

To check what emotions our humor-equipped talking system triggered in the users, we used the Emotive Elements/Emotive Expressions Analysis System (ML-Ask), which determines emotiveness of utterances. The system was developed as a part of a previous project [11]. Based on some ideas presented in our earlier publications, the system performs utterance analysis in two general steps: **1. Determining its general emotiveness** (emotive/non-emotive) and **2. Specifying the types of emotions found** (in emotive utterances only).

In the **first step**, if many of the user's utterances were determined as emotive, it was assumed that he or she was emotionally involved in the dialogue. In Japanese, emotional engagement in the conversation suggests a tendency to familiarize with the partner [12] – which, in this case, is the talking system.

In the **second step**, analysis of the specific emotions showed by the evaluators during the conversation provides us with the information of their feelings towards the interlocutor (in this case – the talking system). If the detected emotions were positive or changing from negative through neutral (non-emotive) to positive during the whole conversation, the general sentiment towards the talking system was considered as

[1] www.goo.ne.jp

positive. If detected emotions are negative or changing from positive through neutral to negative during the whole conversation, the general sentiment towards the talking system was considered as negative.

5 Checking If It Works

We asked 13 people (university students) to perform a conversation with Modalin (without humor, referred to as System A) and with Pundalin (with humor, referred to as System B). As the dialogue was supposed to be as free as possible, no topic restrictions were made. Next, users filled out a questionnaire about the dialogue. Then, the chat logs were checked with the emotiveness analysis system for the presence of different types of emotions.

In previous experiments we asked third person (non-user) human evaluators to tag the emotions in the utterances manually – this, however, is quite laborious and often very subjective, as people tend to annotate emotions with a personal attitude. Using the emotiveness analysis system in the evaluation not only lets us reduce human labor, but also allows us to get more robust and objective results.

6 Did It Work?

6.1 User's Point of View

A summary of the user's questionnaire was given in our previous work. Among the questions asked in this part of the evaluation, 5 are of high relevance to the subject discussed in this paper[2]: 1) Do you want to continue the dialogue? 2) Do you think that the system was human-like? 3) Do you think the system tried to make the dialogue more funny and interesting? 4) Did you find the system's talk interesting and funny? and 5) Which system do you think was better? The first question concerned the willingness to interact with the computer, which obviously is desirable in the field of designing user friendly systems – we can construct extremely sophisticated engines, but it will mean literally nothing if no one would like to interact with them. In the second question we asked directly about the human-likeness of the system. Of course, there is no one and firm definition of this category, and we do not actually aim to find any, as we decided to focus on the user's subjective impression here. The same approach was employed in the following questions, which, albeit quite general, simply give us the answer – which system does the user choose?

The answers to the first four questions were given in a 5-point scale, while for the last one the user only had to choose one of the two systems. Results are shown in Table 1. Statistic significance of all results was checked using the Student's t-test. All differences (apart from question 1. - p value=0.06) were found to be statistically significant on 5% level, which means that the results are robust.

The scores clearly show that users appreciated the humor-equipped system higher than the one without humor.

[2] Other questions concerned the system's linguistic quality, such as grammatical correctness, and are not relevant to the subject of this paper.

Table 1. User's answers for questions: 1) Do you want to continue the dialogue? 2) Do you think that the system was human-like? 3) Do you think the system tried to make the dialogue more funny and interesting? 4) Did you find the system's talk interesting and funny? and 5) Which system do you think was better? Answers were given on a 5-point scale

Questions	1	2	3	4	5
System A	2.62	2.38	1.92	2.46	2 users (15%)
System B	**3.38**	**3.31**	**4.15**	**4.08**	**11 users (85%)**
difference	**0.76**	**0.93**	**2.23**	**1.62**	

6.2 The Emotive Effect

The tendencies shown by the results of emotive analysis are similar to those gained in the human evaluation. Almost all evaluators were more emotionally engaged in the conversations with System B (Pundalin) - see Figure 1.

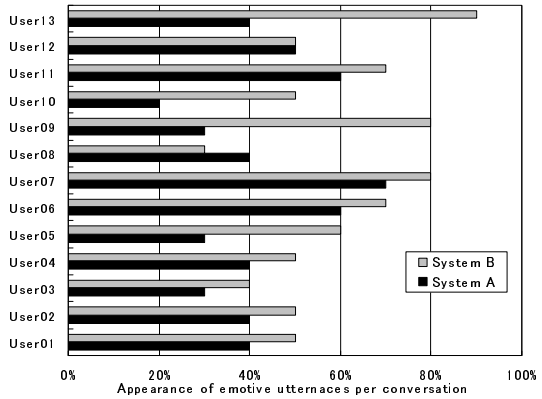


Fig. 1. Average percentage of appearance of emotively engaged utterances for all human evaluators in conversations with System A (without humor) and System B (with humor)

Also the analysis of specified types of emotions clearly showed that evaluators' attitudes to System B were generally positive (78%), while most of the attitudes toward System A (Modalin) were negative (75%) - see Figure 2. The results are discussed in section **7**.

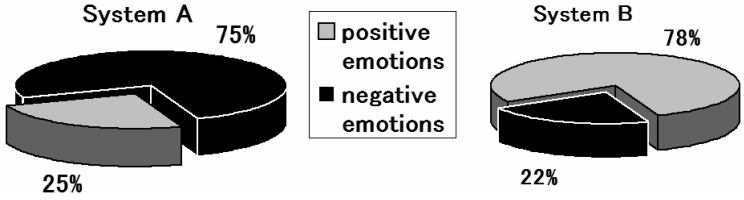


Fig. 2. The total relation of positive emotions to negative conveyed in the utterances of human evaluators with System A (non-humor-equipped) and System B (humor-equipped)

7 What Does It Actually Mean?

The results described above clearly show the supremacy of the humor-equipped system. Of course, the user's point of view should be seen as the most important criterion – it is the user who is going to interact with the system after all. In such understanding, the results of the user experiment are highly satisfying. Increases in human-likeness, willingness to continue the dialogue and overall assessment of the system prove that we have made a step in the right direction.

The results of the user experiment have been discussed in one of our previous works [9]. Here we would like to focus on the emotiveness analysis results and their implications for the field of Humanized Computational Intelligence.

7.1 Affect-as-Information

To understand the results of the emotiveness analysis experiment, we have to take a look at the so-called "affect-as-information" approach, proposed by Schwarz and Clore [13]. The main idea of this approach is based on a claim that humans use affect in the same way as any other criterion, namely – by using the informational value of their affective reactions to form their opinions and judgments. This leads to the assumption that information about someone's attitude to a product can be derived from information about changes in his or her affective states during its usage [14]. Subsequently, according to Sugiyama [12], if we show more emotions towards someone or something, this means that we tend to familiarize with the object.

This leads us to the conclusion that, since the system with humor elicited much more emotions in the users than the one without humor, it can be stated that it is easier to familiarize with it as an interlocutor - which, in fact, is consistent with the claims presented by Sprecher and Regan [5] (mentioned in section 1.2).

7.2 System That Makes Users Feel Better

Thus, we have a humor-equipped talking system that induces more emotions and therefore is more likeable than the other one. This, however, could lead us into a trap: what if most emotions elicited by the system were negative? Obviously, we would acquire a more human-like machine, but not the kind we would like to interact with. This is why we analyzed also the specific types of emotions. The results showed that most emotions induced by the system with humor were positive – and this is what assures us that the effects of our experiment are in fact desirable. In other words, we managed to create a talking system which makes the user feel better. This proves that the beneficial features of humor actually work well also in interaction between humans and machines and should not be neglected in research on user-friendly applications.

8 Emotive-Analysis-Based Evolution of Humor: Adapting to the User's Sense of Humor

Although the experiment described above was a success, obviously many improvements can still be made. The next stage of our research is aimed at creating a

more human-like and more natural joking conversational system that would be able to learn what puns match the preference of the particular user, to recognize when to tell a pun, and to make the user feel better with humor in a more "clever" way than it does now. In other words, the system should be able to adapt to the user. Below we present the outline of our new approach (see sections 8.1, 8.2 and Figure 3).

8.1 Evolving Individualized Sense of Humor

In the experiment described in the previous sections we used an emotiveness analysis system to determine the user's emotional states. However, the applicability of this system is not limited to the evaluation of dialogues – the results of analysis can also be used in a sense of humor evolution algorithm. During the dialogue, the system can analyze each user's utterance, and check how he/she reacts to particular types of jokes. On this basis, the system can build a representation of the user's sense of humor, determining the types of jokes the user likes/dislikes. For example, if the user reacts with positive emotions to jokes concerning politics, the system can assume that this type of joke matches his/her sense of humor. In this manner, the longer the system would talk to the user, the more accurate "tags" of humor sense it can attach – and this, in effect, shall lead to more personalized, more individualized jokes that with a high probability would be appreciated by the user.

8.2 Reacting to User's Emotions

In its current shape, our joking system uses humor at every third turn of the conversation, which obviously has to be changed. Also here we are planning to make use of the emotiveness analysis system, in order to determine the appropriate timing of jokes. We are now conducting experiments to determine which emotional states allow using humor. From previous research we know that humor can be employed to cope with such negative emotions as stress or anxiety. Also the experiments described in this paper showed that humor can alter the emotions from negative to positive. This leads us to the assumption that the emotive analysis system can be used to determine if the user's current emotional state allows for telling him/her a joke.

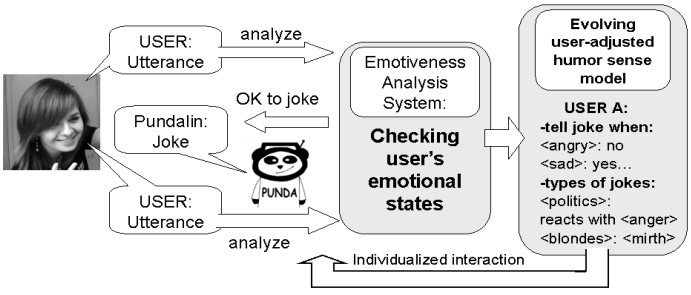


Fig. 3. Outline of the emotive-analysis-based evolution concept (for the humor-equipped conversational system Pundalin)

This idea can be combined with the concept of evolving an individualized humor sense (see section 8.1). Another "tag" in the user's humor sense model can be based on the types of emotions which in the case of this particular user respond well to humor. For example, if the user is angry and does not tend to calm down under the influence of humor, it can be stated that this particular type of emotion does not allow for telling jokes in this user's case. In this manner the system will be able to adjust to the users' needs, which should be highly appreciated by them.

Of course, the amount of tag types is not limited and does not necessarily have to be related to humor. The possibilities given by using the emotiveness analysis system are vast and we believe that this topic should be a subject of thorough discussion.

9 Possible Applications

Final goals of research projects focused on freely talking machines are often misunderstood. Creating a talking-only-system that a user would launch in order to perform a few turns of conversation, is not what we are planning to achieve here. Such systems are only a step towards a more important aim, which is equipping computers (and generally machines we interact with) with a possibility to talk with us in a free (non-task-oriented) way, and, in case of this particular research, to use humor. If we are to treat computers as social actors, they should be able to chat with us and also to use and understand humor, but it does not mean that they should do only that, in a persistent and stubborn way. Therefore, the system presented in this paper should be seen rather as a vessel, created to test some mechanisms and possibilities of creating a functional talking machine. In this meaning, the results we acquired are of high importance and make a good starting point for further research.

As mentioned above, if we are to give chatting and joking possibilities (not obligations!) generally to machines that interact with us, possible applications of such systems are limitless. However, there are some particular cases in which the functions of chatting and using humor are of high importance. There have been some inquiries from car-making companies, interested in creating an intelligent, user-friendly car navigator, able to perform the function of being the driver's conversation partner. Such devices (currently under development) will be able to detect when the driver is getting sleepy, and to do something to wake him/her up. One way of activating driver's brain and preventing him/her from losing attention would be the use of humor. Therefore, implementing a humor-equipped talking system into a car navigator seems like a very promising idea and, in fact, it has already become the subject of car companies' interests.

10 Conclusion

In this paper we investigated the potential of humor in human-computer interaction. The results of our research are consistent with existing literature on this subject and showed that the communication between humans and machines can be significantly enhanced with humor. Automatic emotiveness evaluation showed that after having talked with the joking system users conveyed more positive emotions that in the case

of the non-humorous one. The next phase of our research is aimed at creating a user-adapting system with an individualized humor sense (presented in section 8).

In our research we focus on puns as the most easily computable genre of jokes; however, we believe that some mechanisms developed in this project are applicable also to other types of humor. The results presented in this paper, for example, confirmed one very important hypothesis: even very simple (pun-based) humor can visibly enhance interaction between humans and computers – which obviously should work also for other types of jokes and humor in general. This gives our research more universal dimension and is reason enough to proceed further with this project.

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